

Attracting the Elite

Coordinated Sorting of a Heterogeneous Tourist Population
under an Asymmetric Peer–Congestion Externality

Attracting-the-Elite project

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Abstract

We study J *intrinsically identical* resorts facing a heterogeneous tourist population of two types: rich/high-value (H) and poor/low-value (L). Payoffs are composition-dependent: a visitor's utility falls in the mass of co-visitors of each type, and the central structural assumption is that the externality is *asymmetric*, the rich strongly averse to crowding by the poor while the poor are nearly indifferent to the rich, $\alpha_{HL} \gg \alpha_{LH}$. We ask whether resorts that are otherwise identical can raise their *joint* profit by coordinating on *differentiated* prices that split the population by type, relative to a symmetric baseline in which all resorts post one price. The answer is yes, under a precise condition. We give an exact profit decomposition, $\Pi^S - \Pi^P = \Delta W + T^P - R$, separating three forces: cross-type congestion relief, own-type concentration cost, and the trade of the rent a common price must leave (the pooling rent T^P) for the screening rent R required to separate types. The asymmetry plays a *dual* role: α_{HL} both creates real surplus and relaxes the high type's incentive constraint, which is the screening technology. We also show a countervailing force that informal accounts miss: a *low* α_{LH} makes the poor harder to keep out of premium resorts. At the optimal split, sorting raises total surplus iff $\alpha_{HL} + \alpha_{LH} > 2\sqrt{\alpha_{HH}\alpha_{LL}}$, and the same inequality governs the local instability of the pooled equilibrium, which links the profit result to a wrong-equilibrium trap.

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1 Introduction

Consider five ski resorts on comparable mountains, with comparable lifts, snow, and service. Nothing physical distinguishes them. Yet we routinely observe such clusters *sorting* into a premium destination that serves a wealthy clientele at high prices and budget destinations that serve a price-sensitive mass market. Where does the differentiation come from when the product is identical?

This note isolates one mechanism: an *asymmetric peer externality*. Wealthy tourists dislike sharing a mountain with a large low-budget crowd, whereas low-budget tourists are largely indifferent to the presence of the wealthy, and may even be drawn to it. Under this asymmetry, a group of identical resorts can raise total profit by coordinating on a *menu* of prices that sorts the two types across locations, even though no resort has any intrinsic advantage. The differentiation is created endogenously by the composition of who shows up.

We keep three forces distinct throughout.

- (i) **Surplus creation.** Sorting physically separates the rich from the poor, deleting the cross-congestion cost that the rich bear under mixing. Because the poor barely mind being moved away from the rich, little or no offsetting loss arises, so the relief is a genuine surplus increase.
- (ii) **Screening.** For sorting to be self-enforcing, the rich must *voluntarily* choose the expensive resort over the cheap, crowded one, and the poor must not invade the premium tier. The same distaste α_{HL} that creates surplus also relaxes the high type's incentive constraint. The asymmetry is the screening device.
- (iii) **Rent accounting.** A single posted price must leave a rent to one type (the pooling rent T^P). Sorting recovers T^P but must pay whatever screening rent R is needed to separate the types. The profit comparison nets these against the concentration cost.

Relation to the literature. The model sits at the intersection of three strands. Congestion and population games (Rosenthal, 1973; Sandholm, 2010) and player-specific congestion (Milchtaich, 1996) supply the equilibrium notion but reduce the externality to scalar load. Club-good theory (Buchanan, 1965; Scotchmer, 1994, 2002) captures composition preferences but is static and non-strategic on the provider side. Schelling (1971) has composition thresholds but no prices and no firms. Our contribution is to combine *composition-dependent* payoffs with an *asymmetric* cross-type derivative *and* a strategic (here, colluding) provider layer, and to characterize exactly when type-segregation is profitable and stable.

2 Primitives

Population. There is a nonatomic continuum of tourists of two types $\theta \in \{H, L\}$, with total masses $m_H > 0$ (rich) and $m_L > 0$ (poor); write $M = m_H + m_L$. A type- θ tourist has gross value (budget, willingness to pay) $v_\theta > 0$, with a strict value gap

$$v_H > v_L > 0, \quad d := v_H - v_L. \quad (1)$$

Each tourist chooses one resort or an outside option normalized to utility 0.

Resorts. There are $J \geq 2$ resorts, indexed $j \in \{1, \dots, J\}$, *intrinsically identical*. Resort j posts a price $p_j \geq 0$; marginal cost is normalized to 0. Let $x_{\theta j} \geq 0$ be the mass of type θ at resort j and $x_{\theta 0} \geq 0$ the mass taking the outside option, with adding-up $x_{\theta 0} + \sum_j x_{\theta j} = m_\theta$.

Payoffs. A type- θ tourist at resort j obtains

$$u_{\theta j}(x, p) = v_\theta - p_j - c_\theta(x_{Hj}, x_{Lj}), \quad (2)$$

where the congestion / peer cost is linear in the composition,

$$\begin{pmatrix} c_H \\ c_L \end{pmatrix} = \begin{pmatrix} \alpha_{HH} & \alpha_{HL} \\ \alpha_{LH} & \alpha_{LL} \end{pmatrix} \begin{pmatrix} x_{Hj} \\ x_{Lj} \end{pmatrix}, \quad A = \begin{pmatrix} \alpha_{HH} & \alpha_{HL} \\ \alpha_{LH} & \alpha_{LL} \end{pmatrix}. \quad (3)$$

The cartel maximizes joint profit $\Pi(p, x) = \sum_j p_j (x_{Hj} + x_{Lj})$ and may use transfers across resorts. Because resorts are identical, any asymmetric policy must be sustained by prices and the resulting composition, not by an exogenous quality difference.

Assumption 1 (Asymmetric peer externality). The rich dislike crowding by the poor far more than the reverse, $\alpha_{HL} \gg \alpha_{LH} \geq 0$. The clean benchmark is the one-directional case $\alpha_{LH} = 0$. Own-type crowding is strictly positive, $\alpha_{HH}, \alpha_{LL} > 0$, capturing physical congestion (lift lines, restaurants) and preventing degenerate concentration. The aspirational variant $\alpha_{LH} < 0$ is treated in Section 8.

3 Tourist equilibrium given prices

Fix a price profile $\mathbf{p} = (p_1, \dots, p_J)$. Tourists are followers; resorts are leaders (a Stackelberg structure). Follower play is a population game.

Definition 1 (Wardrop–Nash equilibrium). A feasible allocation x is an equilibrium given $\mathbf{p}$ if for each type θ there is a scalar $\bar{u}_\theta \geq 0$ such that

$$x_{\theta j} > 0 \Rightarrow u_{\theta j}(x, p) = \bar{u}_\theta, \quad x_{\theta j} = 0 \Rightarrow u_{\theta j}(x, p) \leq \bar{u}_\theta, \quad x_{\theta 0} > 0 \Rightarrow \bar{u}_\theta = 0.$$

Each type uses only utility-maximizing resorts and takes the outside option only when no resort yields positive utility.

Remark 1 (Not a potential game in general). Since tourists are nonatomic, a deviator treats resort loads as fixed; with continuous costs an equilibrium exists. The follower game admits an exact potential only when the cross-effects are symmetric, $\alpha_{HL} = \alpha_{LH}$; under Assumption 1 the matrix A is asymmetric, so the game is *not* a potential game and equilibria may be non-unique. We therefore work with explicit allocations and verify equilibrium by the inequalities of Definition 1 rather than by appeal to a potential. When all resorts in a tier share price and composition and $\alpha_{HH}, \alpha_{LL} > 0$, own-type congestion equalizes loads within the tier.

4 Baseline: symmetric pricing (pooling)

Suppose every resort posts the same price p . The symmetric pooled allocation

$$x_{Hj}^P = \frac{m_H}{J}, \quad x_{Lj}^P = \frac{m_L}{J} \quad \text{for all } j \quad (4)$$

is an equilibrium whenever both types participate. Define each type's net (congestion-adjusted) willingness to pay:

$$w_H^P = v_H - \frac{\alpha_{HH}m_H + \alpha_{HL}m_L}{J}, \quad w_L^P = v_L - \frac{\alpha_{LH}m_H + \alpha_{LL}m_L}{J}. \quad (5)$$

A single posted price cannot separate types. Assuming $\min\{w_H^P, w_L^P\} \geq 0$, the largest common price keeping both types in the market is $p^P = \min\{w_H^P, w_L^P\}$, so

$$\boxed{\Pi^P = M p^P = M \min\{w_H^P, w_L^P\}.} \quad (6)$$

Total tourist rent under pooling and total surplus are

$$T^P = m_H(w_H^P - p^P) + m_L(w_L^P - p^P) \geq 0, \quad (7)$$

$$W^P = m_H v_H + m_L v_L - \frac{\alpha_{HH}m_H^2 + (\alpha_{HL} + \alpha_{LH})m_H m_L + \alpha_{LL}m_L^2}{J}, \quad (8)$$

and since marginal cost is zero, $\Pi^P = W^P - T^P$. A common price leaves a positive rent to the type whose net WTP is not the minimum; this recovered rent reappears below. (Equation (6) is the pooling benchmark; a symmetric cartel could instead *exclude* one type, e.g. serve only H at profit $m_H(v_H - \alpha_{HH}m_H/J)$ if that price also deters L . A claim that sorting beats *every* symmetric policy must also dominate these exclusion branches; we return to this in Remark 3.)

5 Coordinated differentiation (separating regime)

The cartel designates $K \in \{1, \dots, J-1\}$ resorts as *premium* at price p_H and $J-K$ resorts as *budget* at price p_L . The intended separating allocation is all rich at premium resorts and all poor at budget resorts, evenly split within each tier:

$$\text{premium: } x_H = \frac{m_H}{K}, \quad x_L = 0; \quad \text{budget: } x_H = 0, \quad x_L = \frac{m_L}{J-K}. \quad (9)$$

Define the four congestion magnitudes

$$h_K = \frac{\alpha_{HH}m_H}{K}, \quad \ell_K = \frac{\alpha_{LL}m_L}{J-K}, \quad z_K^{HL} = \frac{\alpha_{HL}m_L}{J-K}, \quad z_K^{LH} = \frac{\alpha_{LH}m_H}{K}, \quad (10)$$

where h_K, ℓ_K are on-path own-type costs, z_K^{HL} is the cost a rich deviator would incur at a budget resort, and z_K^{LH} the cost a poor deviator would incur at a premium resort.

5.1 Incentive and participation constraints

A deviation is infinitesimal, so it does not change a resort's composition. The separating allocation is an equilibrium iff

$$(\text{IC-H}) \quad p_H + h_K \leq p_L + z_K^{HL}, \quad (11)$$

$$(\text{IC-L}) \quad p_L + \ell_K \leq p_H + z_K^{LH}, \quad (12)$$

$$(\text{IR-H}) \quad p_H + h_K \leq v_H, \quad (13)$$

$$(\text{IR-L}) \quad p_L + \ell_K \leq v_L. \quad (14)$$

Define the *generalized prices* (price plus own congestion borne) and two key gaps,

$$q_H = p_H + h_K, \quad q_L = p_L + \ell_K, \quad A_K = z_K^{HL} - \ell_K = \frac{(\alpha_{HL} - \alpha_{LL})m_L}{J - K}, \quad B_K = h_K - z_K^{LH} = \frac{(\alpha_{HH} - \alpha_{LH})m_H}{K}. \quad (15)$$

The two incentive constraints collapse to the clean sandwich

$$\boxed{B_K \leq q_H - q_L \leq A_K.} \quad (16)$$

A_K is the largest generalized premium the rich will accept over the budget tier; a larger α_{HL} raises A_K , relaxing (11). B_K is the smallest premium that keeps the poor out. Price-only separation for a given K is feasible iff

$$A_K \geq B_K \iff z_K^{HL} + z_K^{LH} \geq h_K + \ell_K. \quad (17)$$

Remark 2 (The overlooked constraint). Strong one-way aversion (large α_{HL}) helps deter the *rich* from the budget tier through A_K , but it does nothing for B_K . Worse, the asymmetry works against (12): a *small* α_{LH} raises $B_K = (\alpha_{HH} - \alpha_{LH})m_H/K$, so the poor find the premium tier relatively attractive and must be priced out. One-way aversion is therefore helpful through α_{HL} but not mechanically helpful for every constraint; feasibility requires checking (17), not merely a large α_{HL} .

5.2 Optimal screening prices for a fixed split

Proposition 1 (Optimal separating prices). *Fix an admissible K (one with $A_K \geq B_K$ and non-negative prices below). The joint-profit-maximizing generalized prices are*

$$(q_H^*, q_L^*) = \begin{cases} (v_L + A_K, v_L), & d > A_K, \\ (v_H, v_L), & B_K \leq d \leq A_K, \\ (v_H, v_H - B_K), & d < B_K, \end{cases} \quad p_H^* = q_H^* - h_K, \quad p_L^* = q_L^* - \ell_K, \quad (18)$$

and the resulting total tourist (screening) rent is

$$\boxed{R_K = m_H (d - A_K)_+ + m_L (B_K - d)_+, \quad (z)_+ = \max\{z, 0\}.} \quad (19)$$

Proof. For fixed K the costs and quantities are constants, so the cartel maximizes $m_H(q_H - h_K) + m_L(q_L - \ell_K)$ subject to $q_H \leq v_H$, $q_L \leq v_L$ and (16). The objective increases in both generalized prices, so push each up to its binding constraint. If $d > A_K$ the two IR constraints cannot both bind (their gap d exceeds the allowed A_K), so $q_L = v_L$ and (11) binds at $q_H = v_L + A_K$. If $B_K \leq d \leq A_K$ both IR bind, $(q_H, q_L) = (v_H, v_L)$. If $d < B_K$ then $q_H = v_H$ and (12) binds at $q_L = v_H - B_K$. Tourist rent is $\sum_\theta m_\theta(v_\theta - q_\theta^*)$, which gives (19). $\square$

Lemma 1 (The asymmetry is the screening device). *In the middle (full-extraction) case $B_K \leq d \leq A_K$ the differentiated prices implement exactly what type-contingent pricing would: resort choice reveals type and the cartel extracts all surplus, $R_K = 0$. The high type's information rent is $m_H(d - A_K)_+$, which falls one for one as α_{HL} rises (through A_K) until it vanishes. This is the precise sense in which asymmetric peer congestion is a screening instrument.*

5.3 Sorted surplus and profit

With cross-type contact eliminated, total surplus and profit are

$$W^S(K) = m_H v_H + m_L v_L - \frac{\alpha_{HH} m_H^2}{K} - \frac{\alpha_{LL} m_L^2}{J-K}, \quad \boxed{\Pi^S(K) = W^S(K) - R_K.} \quad (20)$$

6 Main result: when does sorting beat pooling?

The real-surplus gain from sorting is

$$\Delta W(K) = W^S(K) - W^P = \underbrace{\frac{(\alpha_{HL} + \alpha_{LH}) m_H m_L}{J}}_{\text{cross-type relief}} - \underbrace{\left[\alpha_{HH} m_H^2 \left(\frac{1}{K} - \frac{1}{J} \right) + \alpha_{LL} m_L^2 \left(\frac{1}{J-K} - \frac{1}{J} \right) \right]}_{\text{own-type concentration cost}}. \quad (21)$$

Proposition 2 (Exact sorting-dominance condition). *For any admissible K , coordinated sorting strictly raises joint profit over the symmetric pooling benchmark, $\Pi^S(K) > \Pi^P$, if and only if*

$$\boxed{\Delta W(K) + T^P - R_K > 0.} \quad (22)$$

Proof. $\Pi^P = W^P - T^P$ from (6)–(7) and $\Pi^S(K) = W^S(K) - R_K$ from (20). Subtract: $\Pi^S(K) - \Pi^P = (W^S(K) - W^P) + T^P - R_K = \Delta W(K) + T^P - R_K$. $\square$

Equation (22) is transparent: sorting wins when the congestion relief plus the recovered pooling rent T^P exceeds the own-type concentration cost plus the screening rent R_K it must pay. The asymmetric term α_{HL} enters *twice*: directly in the relief term of ΔW , and indirectly by shrinking R_K through A_K (Lemma 1). It is *not* enough that α_{HL} be large: a severe own-congestion (α_{HH}, α_{LL} large) or an infeasible split ($A_K < B_K$, Remark 2) can still defeat sorting.

6.1 Optimal number of premium resorts

Within the full-extraction region ($R_K = 0$), only the concentration cost depends on K , so the cartel minimizes $C(K) = \alpha_{HH} m_H^2 / K + \alpha_{LL} m_L^2 / (J - K)$.

Corollary 1 (Optimal split and clean threshold). *The interior minimizer is*

$$\frac{K^*}{J} = \frac{\sqrt{\alpha_{HH}} m_H}{\sqrt{\alpha_{HH}} m_H + \sqrt{\alpha_{LL}} m_L}, \quad C(K^*) = \frac{(\sqrt{\alpha_{HH}} m_H + \sqrt{\alpha_{LL}} m_L)^2}{J}, \quad (23)$$

rounded to the nearest admissible integer. If K^ lies in the full-extraction region ($B_{K^*} \leq d \leq A_{K^*}$), then*

$$\boxed{\Pi^S(K^*) - \Pi^P = \frac{(\alpha_{HL} + \alpha_{LH} - 2\sqrt{\alpha_{HH} \alpha_{LL}}) m_H m_L}{J} + T^P,} \quad (24)$$

so sorting strictly dominates pooling iff that expression is positive. Comparing total surplus (ignoring the rent transfer T^P) gives the especially clean criterion

$$\boxed{\alpha_{HL} + \alpha_{LH} > 2\sqrt{\alpha_{HH} \alpha_{LL}}.} \quad (25)$$

Under near one-way aversion ($\alpha_{LH} \approx 0$), a sufficiently large α_{HL} alone satisfies (25). The extra $T^P \geq 0$ shows the cartel can profit from sorting even when sorting slightly *lowers* total surplus, because it also improves price discrimination.

Remark 3 (Dominating exclusion policies). To claim sorting beats *every* symmetric policy, not just pooling, one adds $\Pi^S(K) > m_H(v_H - \alpha_{HH}m_H/J)$ and the analogous L -only inequality. These are easy to check numerically and hold whenever the served mass and value gap are large enough that excluding a whole type is wasteful.

7 The role of the asymmetry

The asymmetry has two distinct roles and one trap.

- (a) **Surplus source.** α_{HL} enters the relief term $(\alpha_{HL} + \alpha_{LH})m_Hm_L/J$ in (21). With $\alpha_{LH} \approx 0$, separating the rich from the poor removes a real loss for the rich without removing a corresponding benefit for the poor.
- (b) **Screening technology.** α_{HL} raises $A_K = (\alpha_{HL} - \alpha_{LL})m_L/(J - K)$, relaxing the high type's incentive constraint (11) and shrinking the rent R_K in (19). The budget tier, crowded with poor, is an unattractive outside option for the rich.
- (c) **The countervailing trap.** A low α_{LH} raises $B_K = (\alpha_{HH} - \alpha_{LH})m_H/K$, the premium the cartel must charge to keep the poor out (Remark 2). If B_K exceeds A_K the separating menu is infeasible at any prices. So the very feature that makes the rich willing to pay can make the poor eager to follow, and both incentive bounds must be checked.

8 What changes or breaks the result

Symmetric externality ($\alpha_{HL} = \alpha_{LH} > 0$). Cross-type contact is mutually costly; sorting yields a larger surplus gain and both types weakly prefer separation. The mechanism is no longer elite avoidance but mutual segregation. The fully homogeneous case $\alpha_{HH} = \alpha_{HL} = \alpha_{LH} = \alpha_{LL}$ gives $A_K = B_K = 0$ at the equal split: resort labels carry no screening content and any apparent gain vanishes once the rent is accounted.

Aspirational poor ($\alpha_{LH} < 0$). If the poor enjoy proximity to the rich, sorting destroys that benefit: the relief term $(\alpha_{HL} + \alpha_{LH})m_Hm_L/J$ shrinks and $B_K = (\alpha_{HH} - \alpha_{LH})m_H/K$ rises, hurting sorting through both the surplus and the IC channels. For sufficiently negative α_{LH} , price-only separation can be infeasible or unprofitable even when the rich strongly dislike the poor. This is the empirically interesting “see and be seen” case.

Large own-type congestion. Sorting concentrates each type into fewer resorts. When α_{HH} or α_{LL} is large the concentration cost in (21) can exceed the cross-type relief; this is the central technological trade-off and is exactly criterion (25).

Failure of price-only screening. If $A_K < B_K$ for every K , no pair of posted prices supports full separation: the price needed to deter the poor pushes the rich to the budget tier. Partial mixing, exclusion, or a second instrument (capacity, branding, minimum spend) is then required.

Cartel enforceability. The analysis maximizes joint profit with transfers. Without transfers, premium and budget resorts earn different profits and may disagree on the policy. Without enforcement, a single resort may deviate to attract both types. The non-cooperative pricing game among resorts is a separate question; notably the asymmetric congestion is self-limiting there, since

a premium resort that cuts price to add volume imports the poor and the very cross-congestion that repels its rich clientele.

9 Dynamics and the wrong-equilibrium trap

The pooled allocation is an equilibrium whenever all resorts post one price and both types participate, *even when* a sorted policy yields strictly higher profit and surplus. Since resorts are identical, moving from pooling to sorting requires a coordinated change in prices or expectations. Consider the within-type replicator dynamic $\dot{y}_{\theta j} = y_{\theta j} (u_{\theta j} - \bar{u}_{\theta})$ with $y_{\theta j} = x_{\theta j}/m_{\theta}$; Wardrop equilibria are its rest points. Perturbing two equal-price resorts by moving masses (δ_H, δ_L) from one to the other, the payoff gap for type θ is $u_{\theta 1} - u_{\theta 2} = -2(\alpha_{\theta H}\delta_H + \alpha_{\theta L}\delta_L)$, so the linearization is governed by $-\text{diag}(m_H, m_L) A$. The pooled state is locally stable iff

$$\boxed{\alpha_{HH} \alpha_{LL} > \alpha_{HL} \alpha_{LH}}. \quad (26)$$

Under strong one-way aversion ($\alpha_{LH} \approx 0$) the right side is near zero, so pooling stays locally stable *however large α_{HL} is*, even though sorting is more profitable and, by (25), more efficient. This is a sharp wrong-equilibrium trap: unilateral tourist adjustment does not create the sorted outcome, and only a coordinated policy jump (price plus, if needed, capacity) crosses the basin boundary. A sorted state with *strict* cross-tier incentives is locally resistant to invasion; when an IC bound binds it is only weakly resistant and small shocks can reintroduce mixing.

10 Extensions

Capacity. A cap k_j changes equilibrium congestion and can ration the premium tier, repairing the invasion failure $A_K < B_K$; but a cap alone does not distinguish types, so it screens only when paired with a price (or auction, minimum spend, priority booking) that selects high WTP.

Branding and amenities. If resort j chooses a brand b_j shifting gross values to $v_{\theta}(b_j)$, a premium brand supports sorting by raising the value gap $d = v_H(b_j) - v_L(b_j)$, moving the economy into the full-extraction window $B_K \leq d \leq A_K$.

Hard budgets. If the poor face a hard budget $\bar{p}_L$, then $p_H > \bar{p}_L$ mechanically blocks invasion, relaxing or removing (12). The present model is more demanding: separation works through preferences, not an exogenous inability to pay.

More than two types. With a continuum of types resorts form a menu of (price, peer-environment) pairs; the two-type sandwich (16) becomes adjacent-type incentive constraints, and a single-crossing condition on type-specific peer costs delivers monotone sorting.

11 Conclusion

Identical resorts, a heterogeneous population, and an asymmetric peer externality ($\alpha_{HL} \gg \alpha_{LH}$) suffice to make coordinated price differentiation strictly more profitable than symmetric pricing, under the exact condition $\Delta W(K) + T^P - R_K > 0$ of Proposition 2. Segregation removes the congestion the rich suffer from the poor; because the poor are nearly indifferent the relief is largely pure surplus, which the cartel captures by charging the rich a premium. The asymmetry's deeper

role is as a screening technology: α_{HL} relaxes the high type's incentive constraint and shrinks its information rent. Yet asymmetry is not a free lunch. A low α_{LH} makes the poor harder to exclude, the concentration cost can dominate, and (criterion (26)) the profitable sorted outcome can be dynamically unreachable from pooling without coordinated policy, the wrong-equilibrium trap. At the optimal split the efficiency of sorting reduces to the single inequality $\alpha_{HL} + \alpha_{LH} > 2\sqrt{\alpha_{HH}\alpha_{LL}}$.

A Notation

Symbol	Meaning
H, L	rich/high-value and poor/low-value tourist types
m_H, m_L, M	mass of each type; total $M = m_H + m_L$
v_H, v_L, d	gross values, $v_H > v_L$; value gap $d = v_H - v_L$
J	number of identical resorts
$K, J - K$	number of premium and budget resorts
p_j	price posted by resort j
$x_{\theta j}$	mass of type θ at resort j
α_{HH}, α_{LL}	own-type (physical) congestion sensitivities
α_{HL}	rich aversion to poor crowding (large)
α_{LH}	poor aversion to rich crowding (≈ 0 , or < 0 aspirational)
h_K, ℓ_K	on-path own-type congestion costs
z_K^{HL}, z_K^{LH}	cross-tier deviation congestion costs
q_H, q_L	generalized prices, $q_\theta = p_\theta + (\text{own congestion})$
A_K, B_K	upper / lower bound on the generalized premium $q_H - q_L$
R_K	screening rent left to tourists under sorting
T^P	rent a single pooling price must leave
$W^\bullet, \Pi^\bullet$	total surplus and joint (cartel) profit